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Steel Building, pre-engineered building, aircraft hangar, IS code 800-2007, air bus A-380.



ABSTRACT

The pre-engineered steel framed structure (PEB) introduced in 1960's includes a structural frame, roofing system and wall supports; for the construction of industrial structure like ware houses, car showrooms, workshops, function halls, stadiums, aircraft hangars etc. is found to be economical, light weight and fast erection process. The PEB's are built to perform some important functions, provide enough space for storing goods; aircrafts etc.. and protecting the storage items against natural disasters, wind forces, and rain. In this research work a hangar for airbus A-380 has been designed considering its dimensions; for maximum span of 120M and width of 115M and a height of 30M as a pre-engineered framed structure. For designing the structural members staad pro (V8i) software is used. This structural design is in accordance with the specification of the general construction in steel confirming to IS:800-2007 (limit state design) and wind application as per latest code confirming to IS:875-2015 (part-III) and earthquake analysis has been done as per IS-1893-2016 (part-IV). For the first time a hangar for A-380 has been designed for using above IS Codes, structural design details, connection details have been produced and found that the total steel is 3502.43 Metric Tons.

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1. INTRODUCTION

In this world of population it is seen that many national, international and domestic airports requires the building for aircrafts servicing. Therefore for servicing and maintaining the aircrafts a building called as a hangar is required so, for the construction of an aircraft hangar more space is required because aircraft hangar consumes more space for its erection and required experienced structural designers. Space for erecting the hangar is the major criteria for some airports where space matters for its maintenance because in an airport other amenities are required to be adjusted for different purposes and need permissions from competent airport authorities otherwise it is highly difficult for aircrafts to be maintained in good condition because in every arrivals to the airports the aircrafts need to get serviced

for smooth travelling and they need to be parked in closed structure, so that it can be safe from whether conditions.

2. LITERATURE REVIEW

(Animesh et al., 2020), conducted a Parametric Study on Design of P.E building Using IS-800-2007 steel code and AISC 360-2010, 13th edition.

(Nihar shah et al., 2021), did a comparative study of PEB by designing a P.E.B & C.S.B using IS-codes & American codes and found that the design of PEB and CSB structure using an AISC code gives 3% to 10% light weight sections as compared to IS-codes. AISC code has less factor of safety compared to IS-code. Being a less factor of safety, AISC code

resulting a light weight sections which is not suitable for india to erect the pre-engineered building.

3. METHODOLOGY

This study includes the designing of an aircraft hangar for airbus A-380 as a pre-engineered steel framed building using indian standard code (IS-code) willing to erect this structure in future if the airbus A-380 is bring to india as a own indian aircraft for indians travelling from own country to other countries

4. PRE-ENGINEERED BUILDING (PEB)

Pre-engineered buildings are the structures that are constructed for the purpose of work shops,garriages, storage rooms,servicing centres etc. For constructing these structures we required designing software and based on the customers need all the members of the frame are desined.(Anisha Goswami et al., 2018).

All the components are made in steel plant itself for structural and aesthetic requirements. These are tapered sections which are combinations of Built-up section. (Saranya and Shallini 2019)Manufacturing process like cutting, welding and painting is done at the plant itself.Manufactured materials are transported to customer site under completely knocked down condition(CKD). The pre-engineered building members are erected at site through high strength grip nut and bolts system. Under same loading Conditions these are shifted to the selected place. Restoration of buildings happens in case of Pre Engineering system. PEB is entirely a concept of Superstructure. In pre-engineered building the main Member is tapered section and Secondary members are "Z" and "C" section. Delivery of all members takes 6 to 8 weeks approximately; Foundation design is very simple for these types of buildings. Time for Erection and estimated budget will be known before construction. In PEB installation procedure is easy, fast, with the help of lifting machines. Majority of pre-engineered building are used due to low weight and offer higher stresses to natural forces. (ShahidWasim Chaudhary et al., 2019)

The pre-engineered building system can be applied to the following steel structures.

- Industrial Buildings
 - Ware Houses
 - Air Craft Hangars
 - Commercial Complexes
 - Celebration halls
 - School Buildings
 - Metro Stations
 - Parking Lots
 - Stadiums
 - Railway Platforms
- (Anisha Goswami et al., 2018).

4.1 Aircraft hangar

An aircraft hangar is a closed structure which is constructed for the use of servicing and maintenance of various aircraft as a protective space because whenever an aircraft arrival be seen after a long journey it has to be taken under that hangar where all the equipment's are available for it servicing. In an aircraft hangar more space is required for its storage and extra space for its maintenance. Majority of the hangars are design as a pre-engineered steel building due to its light weight and higher resistance over wind and seismic loads.

5. CALCULATIONS

Primary Loads

D.L:self-weight

L.L: imposed load

WL: W.L with internal coefficient +/- 0.18

E.L: Seismic Load

Dead Load:

Dead load has been considered as 0.1 Kn/m² due to weight of sheeting+Purlins and Roof insulation +Self weight of frame.

Live Load:

Live load on roof has been considered as - 0.75 Kn/M²

Collateral loads:

Light load – 0.15kn/m

Wind loads:

V = 44 m/sec,

K1 = 1.00(for 50years)

K2 = 1.095 as per IS 875(Part-III)-2015

K3 = 1.0

K4 =1.0

$V_z = V_b \times K_1 \times K_2 \times K_3$

$P_z = 0.6(V_z)^2 = 1.4 \text{ Kn/m}^2$.

$P_d = K_a \times K_d \times K_c \times P_z$

$P_d = 0.8 \times 0.9 \times 0.9 \times 1.4$

$P_d = 0.90 \text{ KN/M}^2$

Internal pressure coefficient = +/-0.5

Wind application on staad members for High bay

$H/W = 25/120 = 0.209$

$L/W = 115/120 = 0.959$

Wind Speed = 44 m/sec

K1 = 1

K2 = 1.095

K3 = 1

$V_z = 44 \times 1 \times 1.095 \times 1$

$= 48.18 \text{ m/sec}$

$P_z = 0.6 \times V_z^2 = 1393/1000 = 0.908 \text{ Kn/m}^2$

C1 = 0.7

C2 = -0.2

C3 = -0.95

C4 = -0.4

Cpi = 0.5

Ka =0.8

Kd = 0.9

$K_c = 0.9$

Design Wind Pressure

$P_d = K_{ax}K_{dx}K_{cx}P_z$

$P_d = 0.8 \times 0.9 \times 0.9 \times 1.4$

$P_d = 0.908 \text{ kN/m}^2$

Earthquake load:

Earthquake load as per IS 1893(Part-IV) – 2016

Zone-II = 0.10

Imp. factor = 1.5

Res. Red. factor = 4.0

Minimum Thickness Criteria

Secondary Members: 2.5 mm thick

Serviceability Criteria

Main Frame

Vertical deflection : Span / 150

Lateral deflection : Height / 100

Purlins & Girts : Span / 150

6. ANALYSIS AND DESIGN

12.1 Dead Load

$= 0.1 \text{ kN/m}^2$

$= 0.1 \times 7.1875$

$= 0.719 \text{ kN/m}$

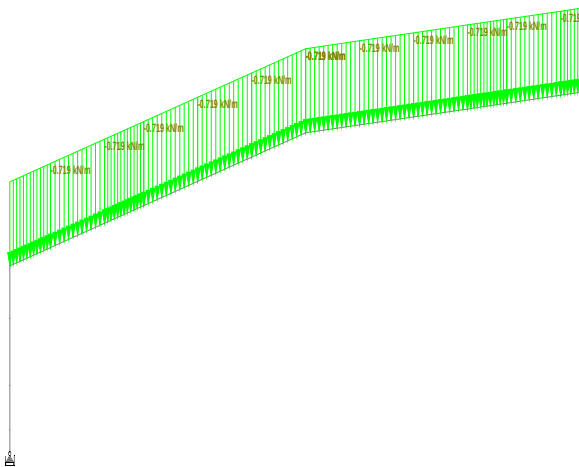


Figure 1. Dead loading diagram

12.2 Live Load

$= 0.75 \text{ kN/m}^2$

$= 0.75 \times 7.1875$

$= 5.391 \text{ kN/m}$

12.3 Collateral load

$= 0.15 \text{ kN/m}^2$

$= 0.15 \times 7.1875$

$= 1.07 \text{ kN/m}$

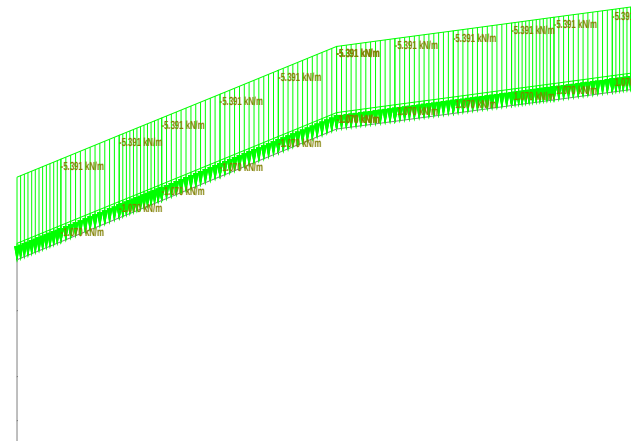


Figure 2. Live loading diagram

13. NODE DISPLACEMENT SUMMARY

The below is the node displacement summary in the table form.

			Summary						
	Node	L/C	Horizontal X mm	Vertical Y mm	Horizontal Z mm	Resultant mm	Rotational		
							rX rad	rY rad	rZ rad
Max X	257	122	109.838	-0.832	-1.437	109.850	-0.000	0.002	0.001
Min X	253	123	-109.848	-0.832	-1.438	109.861	-0.000	-0.002	-0.001
Max Y	70	103	-12.758	77.912	-103.845	130.448	-0.011	-0.000	0.000
Min Y	233	101	0.002	-643.827	72.669	647.915	0.009	0.000	0.000
Max Z	799	102	12.007	-0.107	271.860	272.125	-0.003	0.000	-0.000
Min Z	799	109	-0.576	-0.602	-284.610	284.611	0.006	-0.001	0.000
Max rX	22	124	-0.554	-1.779	112.075	112.091	0.035	0.000	-0.000
Min rX	840	109	0.000	0.000	0.000	0.000	-0.029	0.008	-0.000
Max rY	848	109	0.000	0.000	0.000	0.000	-0.025	0.022	0.000
Min rY	844	109	0.000	0.000	0.000	0.000	-0.025	-0.022	-0.000
Max rZ	877	101	-20.656	-39.977	31.982	55.206	-0.019	-0.003	0.043
Min rZ	886	101	20.730	-39.980	31.973	55.231	-0.019	0.003	-0.043
Max Rst	233	101	0.002	-643.827	72.669	647.915	0.009	0.000	0.000

Table 1 Displacement summary

13.1 Maximum deflection=643.827mm (vertical).

Limiting deflection (V/180) = 120.0m/180.

= 666.66mm

Hence safe in Deflection.

13.2 Maximum deflection=109.838mm

(horizontal).

Limiting deflection (H/150)= 25.0m/150.

=166.66mm

Hence safe in Deflection.

16. RESULTS AND DISCUSSIONS

The structural analysis and design has been done for airbus A-380 hangar as pre-engineered steel framed building by considering the maximum dimensions as 120M x 115M x 26m eave height and 31M clear height as from floor finish level to the top of the frame. For designing the steel frame pre-engineered building hangar staad-pro software has been used. In this design 3d analysis has been done and in the above all the

structural details and drawings have been mentioned. The following results have been obtained from this design.

Table 4.Software analysis results and summary.

Maximum Displacements in mm				Rotational Displacements in radians		
S. No	X-axis	Y- axis	Z- axis	X- axis(Rx)	Y- axis(Ry)	Z- axis(Rz)
1.	109.838 mm	-0.832mm	-1.437mm	-0.000 radians	0.002 radians	0.001 radians
Max. S.F in KN				Max. B.M in KN-M		
S. No	X- axis	Y- axis	Z- axis	X- axis(Mx)	Y- axis(My)	Z- axis(Mz)
2.	3566.697 KN	287.432KN	-3.530KN	-0.672KN-M	28.839KN-M	846.388KN-M
Estimated Quantity of Steel in Metric Tonne						
S. No	Primary Members	Flange Bracings, Sag Rods And CHS/SHS	Secondary Members	Roof sheeting And Wall sheeting	Anchor Bolts and High Strength Bolts	Total Quantity of Steel Obtained
3.	2861.87MT	313.51MT	133.50MT	145.81MT	47.74MT	3502.43MT

17. CONCLUSIONS

Analysis and design in this study yielded the following conclusions:

- The structure designed in this research

Acknowledgement:

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